

Adversarial Learning for Robust Detection of Malicious Ethereum Contracts and Transactions

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Abstract

State-of-the-art machine-learning detectors for malicious smart contracts and transactions (including foundation models and temporal graph networks) remain brittle to carefully crafted adversarial perturbations—subtle bytecode obfuscations, transaction reordering, or feature manipulations—that enable evasion while preserving semantic correctness. This project develops a unified adversarial-training framework that integrates projected-gradient-descent attacks during both contract-embedding and transaction-stream training, producing robust models with provable guarantees under bounded perturbations. Building upon our team’s prior work on foundation models, real-time detection, and analyzer stress-testing [2, 3, 4, 5, 6, 7], the approach advances recent empirical demonstrations of single-feature and semantic adversarial attacks on Ethereum classifiers [8, 9]. We outline min-max optimization objectives, evaluation under white-box and black-box threat models, open-source robust checkpoints, and collaborations in adversarial ML and blockchain deployment.

Problem Formalization

Existing ML-based detectors for malicious contracts and transactions are limited by their vulnerability to adversarial examples: small, semantically-preserving perturbations (e.g., dead-code insertion in bytecode or subtle reordering of transaction features) cause dramatic drops in detection performance, as recently demonstrated on Ethereum phishing and smart-contract classifiers. This project solves these limitations by embedding adversarial training directly into the model optimization pipeline—using projected gradient descent inside a min-max objective—while leveraging the multimodal embeddings and temporal graph representations developed in our prior foundation-model and real-time detection projects, thereby producing models that maintain high accuracy even under strong white-box and black-box attacks.

The proliferation of adversarial machine-learning techniques in blockchain security has exposed a critical weakness: detectors trained on clean data fail catastrophically when attackers apply minimal perturbations that preserve contract functionality or transaction validity. We therefore propose *AdvEVM-Robust*, a unified adversarial-learning framework that jointly hardens both contract-level (bytecode + trace) and transaction-level (mempool stream) models. The core training objective follows the classic min-max saddle-point formulation:

$$\min_{\theta} \max_{\|\Delta\| \leq \epsilon} \mathbb{E}_{(x,y) \sim \mathcal{D}} [\ell(f_{\theta}(x + \Delta), y)],$$

where x is either a bytecode/trace embedding (from Project1) or a transaction-window graph (from Project2), Δ is a perturbation constrained by an ℓ_{∞} or ℓ_2 ball of radius ϵ , and ℓ is the task loss (cross-entropy for classification plus semantic-consistency regularizers).

For contract detection we extend the multimodal EVM-FM architecture with an inner-loop PGD attacker that perturbs tokenized bytecode, CFG edge weights, and trace sequences while preserving EVM semantics (verified via lightweight symbolic simulation). For transaction streams we augment the TGNN with adversarial message-passing: at each time step the attacker injects bounded perturbations into node features (balance deltas, storage writes) or edge timings within mempool constraints. Both pipelines share a common robustness regularizer that aligns clean and adversarial embeddings:

$$\mathcal{L}_{\text{rob}} = \mathbb{E}[d(\mathbf{z}(x), \mathbf{z}(x + \Delta))] + \lambda \cdot \mathcal{L}_{\text{cls}}(f_{\theta}(x + \Delta)),$$

where d is a cosine or KL divergence that enforces representation invariance.

Architecturally, we introduce a lightweight adversarial generator (a small conditional GAN or diffusion model) that produces realistic perturbations on-the-fly during training, eliminating the need for expensive pre-generation of adversarial datasets. The generator is conditioned on the current victim model parameters and is updated via alternating gradient steps, ensuring fresh attacks at every epoch. Formal convergence is guaranteed by standard results in adversarial training under Lipschitz-continuous networks and bounded perturbation budgets.

Evaluation proceeds along three axes: (1) clean accuracy versus state-of-the-art detectors, (2) robustness under increasing perturbation budgets ϵ measured by attack success rate (ASR) and certified robustness radius via randomized smoothing or abstract interpretation, and (3) transferability across threat models (white-box, black-box, and adaptive adversaries that know the defense). We further quantify real-world impact by replaying known DeFi exploits with adversarially hardened detection models on historical Ethereum mainnet traces, measuring reduction in successful evasion and economic loss.

The project delivers open-source robust model checkpoints, an adversarial benchmark suite extending our prior analyzer stress-testing work [7], and a public leaderboard that ranks detectors under realistic attack scenarios. We invite collaborators with expertise in certified robustness, generative modeling, and on-chain deployment (relayers, validators) to co-develop production-grade hardened detectors and to explore extensions to other EVM-compatible chains and Layer-2 rollups. Funded PhD positions and industry partnerships for compute and trace data are particularly welcome.

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